



Evaluation of post-farm-gate passive surveillance in swine for the detection of foot and mouth disease in Australia

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ABSTRACT

Pigs are considered high risk for the introduction and spread of foot and mouth disease (FMD) in Australia. One of the most likely pathways of introduction of FMD into Australia would be through the illegal importation of FMD-contaminated meat, which is then fed to feral or domestic pigs. Locations where animals from different origins are commingled, such as livestock markets and abattoirs, pose a risk for disease spread. Early detection of exotic diseases at these locations is crucial in limiting the spread of an outbreak. The aims of this study were to evaluate the likelihood of exotic disease detection with current passive disease surveillance activities for pigs at saleyards and abattoirs in eastern Australia, and make recommendations for improving surveillance. Sensitivity (Se) of the current post-farm-gate passive surveillance for detection of exotic diseases was estimated using the scenario tree modelling methodology (Martin et al., 2007a). Four surveillance system components were identified: (i) domestic saleyard, (ii) export saleyard, (iii) domestic abattoir, and (iv) export abattoir. Pig farms were classified according to herd size (Small vs. Large) and subsequently into two risk categories depending on the probability of swill feeding (Swill feeding vs. Not swill feeding). A scenario tree representing the pathways by which infected animals could be detected was developed and the Se of detection in each surveillance system component was estimated. Industry statistics, information on previous exotic disease outbreaks, and interviews with pig producers were used to estimate herd category proportions and the relative risk of swill feeding. Quantitative estimates for probabilities of detection were sourced from State legislation and policies, stakeholder consultation and observational studies at saleyards and abattoirs. Results of a FMD case study showed a Se of detection at a representative location for each surveillance system component during a 2-week period of 0.19 at domestic saleyards, 0.40 at export saleyards, 0.32 at domestic abattoirs and, 0.53 at export abattoirs. This output assumed the country was infected with herd and unit design prevalences of 1% and 30%, respectively. Improving disease awareness of saleyard and abattoir stockmen, increasing the presence of inspectors at these venues and identifying those herds posing a higher risk for FMD introduction, could improve the capacity of the country for early detection of emerging animal diseases.

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1. Introduction

The animal health surveillance system of a country generates information on the health and disease status of the nation's animal population (OIE, 2007; Hadorn and Stärk, 2008) that is used to report, assess and manage the risks and impacts of animal diseases upon international trade, public health, animal production and animal welfare (WHO, 1997). Information generated through a surveillance system can be based on both, active and passive surveillance approaches. While active surveillance is defined as the collection of samples by the veterinary health authorities for gathering information on specific diseases in the population, passive surveillance is defined as the reporting of clinical suspect cases to the health authorities by animal owners, practitioners or inspectors (Doherr and Audigé, 2001; Hadorn and Stärk, 2008).

A critical function of a national surveillance system is the early detection of exotic, re-emerging or emerging animal diseases. Early detection underpins a rapid response by the country's relevant authorities and thereby is a critical determinant of the ability to contain or eradicate the disease (Salman et al., 2003). Livestock markets (saleyards) offer an opportunity to inspect animals from multiple sources, to detect abnormalities and to trace animal movements. Saleyards in Australia also provide an opportunity to examine pigs from small-scale pig producers, defined as herds with less than 100 sows, and backyard pig operations. These types of production systems are believed to pose a biosecurity risk to the mainstream pig industry through disease introduction due to their biosecurity practices (perceived as poor), their use of saleyards to trade pigs, and the assumption they are more likely to feed swill (Schembri et al., 2006; Schembri, 2008; Van Metre et al., 2009). Ante-mortem inspection of animals at abattoirs also provides an opportunity for the detection of exotic diseases such as foot and mouth disease (FMD). The importance of ante-mortem inspection at abattoirs was highlighted by the 2001 FMD outbreak in the United Kingdom, which came to the attention of officials by a veterinarian after routine ante-mortem checks on pigs at an abattoir (Bourn, 2002). Previous research (Schemann et al., 2010) evaluating disease surveillance activities at pig saleyards and abattoirs in Australia identified infrequent inspections of the animals and non-uniform inspection methods as gaps in surveillance that could compromise early detection of diseases such as FMD.

Given the importance of abattoir and saleyard surveillance activities in the early detection of FMD it is essential that the performance of surveillance system at saleyards and abattoirs be quantified. Historically, performance estimates for surveillance systems included uncertainty around the prevalence, upper confidence limit if no disease was detected, and likelihood of detecting disease. The methods for estimating these values are based on an assumption that data were collected using structured probabilistic sampling methods (Cannon and Roe, 1982; Cameron and Baldock, 1998). Surveillance undertaken at saleyards and abattoirs cannot be evaluated using these methods because the activities form part of a passive surveillance system rather than a structured survey. The

sensitivity of passive surveillance systems is difficult to assess as it depends on many factors, such as the probability of infected animals showing clinical signs, the disease awareness of farmers, practitioners and inspectors, and their motivation to report (Hadorn and Stärk, 2008). In the case of saleyards, the activities undertaken by government stock inspectors focus on assessing compliance with traceability and welfare legislation rather than having a requirement to inspect a specific number of animals for exotic disease detection. Similarly, ante-mortem inspection at domestic and export abattoirs focus on ensuring animals are fit for human consumption rather than conducting active surveillance for specific diseases. Recently, the stochastic scenario tree methodology has been developed to evaluate abattoir-based serology for Classical Swine Fever in Denmark, taking into consideration the differential risk of sero-positivity associated with age and location (Martin et al., 2007b) and to create a risk-based surveillance system for *Trichinella* spp. in Danish pigs (Alban et al., 2008). This methodology has also been used to evaluate surveillance strategies for demonstrating farmed deer are free of tuberculosis (More et al., 2009).

The scenario tree methodology can also provide a means for evaluating passive surveillance systems but as yet this has not been demonstrated in the literature. The objective of this paper was to utilise stochastic simulation trees to evaluate the passive surveillance activities for FMD at Australian swine saleyards and abattoirs. Specifically, we quantitatively evaluate the likelihood of FMD detection with the current disease surveillance activities for pigs at saleyards and abattoirs in Australia using scenario tree modelling. This will allow identification of potential weaknesses and improvements of the current post-farm-gate passive disease surveillance for pigs in Australia.

2. Methods

Sensitivity (Se) of passive surveillance at a post-farm-gate level (i.e. during activities at livestock markets and abattoirs) for the pig industry in Australia was estimated using the scenario tree methodology described by Martin et al. (2007a) and a FMD outbreak case study.

The scenario tree is a structure representing all possible pathways from the infected population to the detection or not detection of the infection at the surveillance system evaluated. The initial step involved identifying and describing the different components of the post-farm-gate surveillance system. This process identified four surveillance system components (SSC): (i) domestic saleyard, (ii) export saleyard, (iii) domestic abattoir, and (iv) export abattoir. A saleyard was considered domestic when supplying domestic markets only, while an export saleyard supplied domestic and export markets. Pigs at domestic saleyards are mainly sold to re-stocker farms as replacement breeding stock, to be grown out to heavier weights, or for home consumption. A proportion of pigs sold at these saleyards are afterwards slaughtered at domestic abattoirs. At export saleyards, most pigs are bought by butchers or processors and subsequently slaughtered at domestic or export abattoirs (Cutler and Holyoake, 2007).

As a consequence, saleyard and abattoir surveillance system components cannot be considered independent.

A scenario tree representing the pathways and events where infected animals could be detected, with the corresponding specific probabilities, was developed. This scenario tree modelled the probability of FMD being present and detected in a pig being processed through the SSC, in a 2-week period, given the population is infected at specific design prevalences. Sensitivity of detection was estimated during a 2-week period, due to the rapidity of spread of FMD, and among pig herds supplying to a single representative location for each surveillance system component, as a FMD outbreak is likely to be detected before the disease spreads to other regions. Sensitivity of a representative location for each surveillance system component, defined as the probability of detecting at least one positive herd supplying to a single saleyard or abattoir, was calculated. The reference population for calculating the *Se* at a representative location for each surveillance system component (*Location Se*) was the total number of suppliers for each location. The sensitivity of the surveillance system component (*Component Se*) was calculated considering all saleyards, and all abattoirs in the country and the Australian pig population as the reference population. Data required to populate the models, such as population proportions and probabilities of inspection and detection, were determined in a number of ways as described below. Data for the herds and pigs sold to saleyards and abattoirs for the 2-week period were simulated.

The scenario tree was implemented in Microsoft Excel (PC/Windows XP, 2006) and sensitivities were calculated with stochastic simulation using the software @RISK 5.5 (Palisade Corporation, USA). Each simulation comprised 3000 iterations sampled using the Latin hypercube method with a fixed random seed of one.

2.1. Data sources

Results from the following data gathering activities formed the sources of parameter estimates for the analytical model in this assessment, except where other sources are specified.

2.1.1. Disease surveillance legislation and policy review

Legislation related to disease surveillance in each state of Australia was consulted to determine what inspection of livestock is required at saleyards and abattoirs. Legislation was sourced from the Australian Legal Information Institute web page (www.austlii.edu.au) and from key persons within the relevant state government departments. Information on policies and procedures related to disease surveillance was gathered using a questionnaire and consultation with relevant state governments. A review of animal welfare legislation was also included because some diseased animals may present as welfare cases (e.g. foot lesions may cause lameness).

2.1.2. Saleyard and abattoir stakeholder consultation

Two face-to-face group discussions were conducted at the end of 2007 to gather information on post-farm-gate disease surveillance resources and activities at saleyards

and at abattoirs. Government and industry representatives participated at the saleyard group discussion. Abattoir managers, abattoir meat inspectors, transporters, industry representatives and representatives of the Australian Quarantine and Inspection Service (AQIS) and the New South Wales (NSW) Food Authority participated at the abattoir group discussion. Information regarding resources available for disease surveillance among government agencies, as well as current surveillance activities at saleyards and abattoirs, was collected to identify gaps present in the current system and possible alternative systems for improving surveillance.

2.1.3. Observational studies

In 2008, observational studies to evaluate actual surveillance activities at saleyards and abattoirs were conducted in NSW (Schemann et al., 2010), Victoria and Queensland. Quantitative (frequency) and qualitative (methodology) information on disease surveillance activities undertaken by field inspectors, saleyard agents, abattoir ante-mortem inspectors and AQIS inspectors, as well as the incidence of pig health and welfare abnormalities, were recorded.

2.1.4. Producer interviews and discussion groups

Interviews with pig producers were conducted at six saleyards where regular pig sales were held, two in each of the three eastern states of Australia (New South Wales, Victoria and Queensland) (Hernández-Jover et al., 2008; Schembri, 2008; Schembri et al., 2010a). Pig producers who participated in the interviews ($n=106$) were volunteers recruited either from a cohort of producers who responded to a postal survey on demographics in 2006 (Schembri, 2008), or opportunistically during visits to each study saleyard. The interview gathered information on demographics, husbandry practices, nutrition, herd health, biosecurity practices, movement practices and animal identification systems. Between April and July 2007, nine one-off discussion groups were conducted on the day of the pig sale at four saleyards in eastern Australia. The 5–12 producers in each discussion group were volunteers recruited from those interviewed. Topics of discussion were disease knowledge, reporting practices, communication methods, and on-farm and post-farm-gate identification practices (Hernández-Jover et al., 2008, 2009; Schembri, 2008).

2.1.5. Case study interviews and questionnaires

A total of 13 small scale (≤ 100 sows) pig producers selling by informal means (internet, word-of-mouth, family and friends and local businesses) in NSW were interviewed to qualitatively evaluate the potential risk of introduction and spread of FMD among producers trading pigs informally (Hernández-Jover et al., 2010; Schembri et al., 2010b). Supporting information on these producers was collected through a questionnaire with members of the Australian Pig Breeders Association ($n=29$) and among participants ($n=24$) of pig industry field days. These studies collated information on demographics and practices on husbandry, feeding, health management, biosecurity and pig movements.

2.1.6. Australian Pork Limited statistics

Statistics summarizing the pig industry in Australia were obtained from the Australian Pig Annual 2006–2008 (Australian Pork Limited; [APL, 2008](#)). Information used included the 2007 figures for total number of pig establishments (farms) and number in each herd size category (based on the number of breeding sows) for each individual state and for the national herd.

2.1.7. Previous saleyard and abattoir audits

Compliance with a new traceability and food safety requirements for the pig industry in NSW, Victoria and Queensland was evaluated by [Hernández-Jover et al. \(2009\)](#). A benchmarking exercise at saleyards and abattoirs during 2008 was used to investigate implementation, and barriers to adoption, of a national vendor declaration and an on-farm quality assurance program for piggeries, among pig producers in Australia.

2.2. Scenario tree

Sensitivity of detection was estimated for each SSC with the scenario tree illustrated in [Fig. 1](#). This scenario tree models the probability of detecting disease (*i.e.* a positive surveillance outcome) given disease is present at the specified design prevalence. The nodes for the scenario tree are described below and summarized in [Table 1](#). Input values were modified as necessary for each SSC; however the HERD STATUS, ANIMAL STATUS and CLINICAL SIGNS nodes had the same input values for all four SSCs.

2.2.1. FARM TYPE

Pig farms were classified according to herd size: *Small herds* (≤ 100 sows) and *Large herds* ([APL, 2008](#)). Proportions of farms selling through saleyards and abattoirs which fell into the small and large categories were estimated from previous saleyard and abattoir benchmarking audits ([Hernández-Jover et al., 2009](#)). Vendors selling more than 100 animals in a period of 2 weeks were considered to have a herd size of more than 100 sows, assuming a productivity of 20.73 piglets/sow/year ([APL, 2008](#)).

2.2.2. SWILL FEEDING

Farms were also classified by whether or not swill is fed to the pigs. The practice of swill feeding is a risk factor for infection, and the SWILL FEEDING risk node accounted in the analysis for the resulting differential risks of herds being infected, while using a single value for design prevalence across the population ([Martin et al., 2007a](#)). The two branches of this node were *Swill feeding* and *Not swill feeding*. Model parameters that must be estimated for incorporation of the differential risk associated with swill feeding are:

- proportions of pigs processed (inspected) from *Swill feeding* and *Not swill feeding* herds in the SSCs under consideration,
- the risk of FMD infection in *Swill feeding* herds relative to that in *Not swill feeding* herds, and

- proportions of pig producers who are feeding swill in the reference population defined by the model.

2.2.2.1. Proportion of herds processed in the SSC that are feeding swill. Little information is available on the rates of swill feeding among *Large* and *Small herds* processed in each surveillance system component, and values used in this evaluation were based on producers' feeding practices which could potentially lead to swill feeding. Twenty six of 131 small-scale producers participating in data gathering activities were feeding their pigs retail waste and table scraps, while 5 of the 40 large-scale producers reported these practices ([Hernández-Jover et al., 2010](#); [Schembri, 2008](#); [Schembri et al., 2010a](#)). The proportions of *Small herds* and *Large herds* likely to feed swill were included in the model as Pert distributions, with the proportion of producers using retail waste and table scraps to feed their pigs as the maximum value of the distribution, minus 20% and 50% as the most likely and minimum values, respectively.

2.2.2.2. Relative risk Swill feeding: Not swill feeding. The link between feeding waste and FMD was demonstrated during the 2000 outbreak in South Africa ([Brückner et al., 2002](#)) and the 2001 outbreak in the United Kingdom ([DEFRA, 2002](#); [Bourn, 2002](#)), which originated in piggeries following illegal feeding of contaminated untreated swill. In our analysis, the risk of FMD infection in a *Swill feeding* herd (RR_{SF}) relative to the risk in a *Not swill feeding* herd, was based on expert opinion. According to three veterinary epidemiologists and a pig veterinarian, the RR_{SF} was estimated with the @RISK function RiskPert(5,10,20). *Not swill feeding* was the reference category (relative risk = 1) for this risk factor. As explained in the next section, for this assessment the country is assumed to be infected at specific design prevalences, representing there is a level of FMD infection in the country. As a consequence, the estimated RR_{SF} in *Swill feeding* herds is not as high as if the assessment was evaluating the risk of FMD introduction. If infection is present in the country, *Swill feeding* herds can be infected through swill feeding or through spread of FMD from other properties, while *Not swill feeding* herds can only be infected through spread of FMD from other properties.

2.2.2.3. Proportion of herds feeding swill in the reference population. The proportion of producers swill feeding in the reference population was based on the proportion of *Small* and *Large herds* and producers *Swill feeding* and *Not swill feeding* in Australia. The proportion of small and large herds was based on the 2006–2008 industry statistics ([APL, 2008](#)), which information is sourced from the Australian Bureau of Statistics ([ABS, 2009](#)). According to these statistics, there were 1409 (74.4%) small pig herds (≤ 100 sows) and 486 (25.6%) large pig herds in Australia. Previous studies ([Schembri, 2008](#)) indicated that a significant proportion of small pig producers were not registered within the pig industry body (Australian Pork Limited). Moreover, the number of pig herds was based on information obtained from a sample of the agricultural business population ([ABS, 2009](#)). As a consequence, the number of small herds was modelled with a Pert distribution, adding 10% and 15% to the number of small herds stated in the

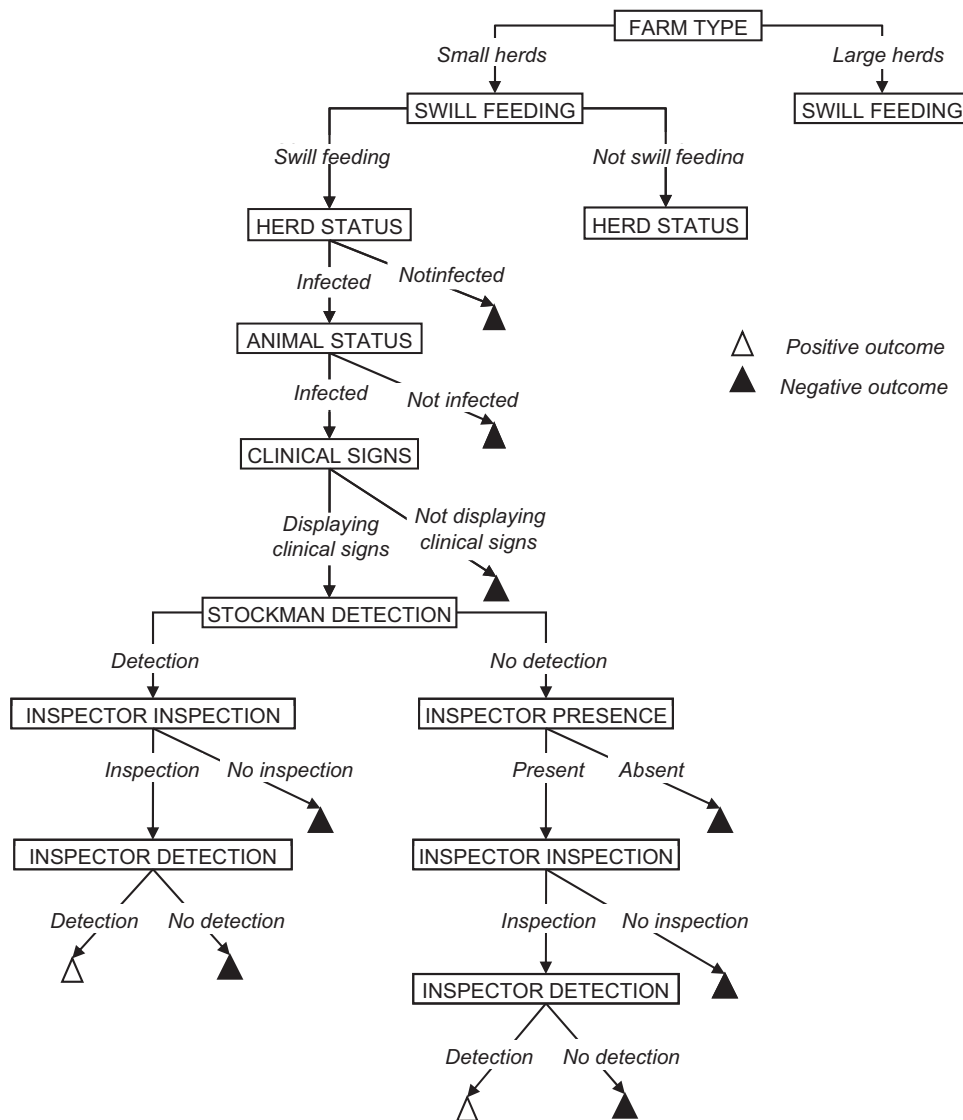


Fig. 1. Scenario tree describing the saleyard and abattoir surveillance system components in Australia. Only nodes for the *Small herds Swill feeding* are shown (the same nodes have been considered for the *Small herds Not swill feeding* branch and the *Large herds* branch).

statistics (RiskPert(1409,1550,1620)). No uncertainty was added to the number of large herds, which was included in the model as a fixed value.

Among 148 swill feeding inspections conducted in Australia during 2006, nine resulted in warning letters for non-compliance with swill feeding legislation. Swill feeding inspections were conducted during swill feeding awareness campaigns, following allegations of swill feeding or during targeted visits to farms with previous history of swill feeding. As the number of inspections only represented 6.2% of the national herd and inspections were not conducted randomly among all pig producers in Australia, the proportion of producers feeding swill was modelled using the @RISK function Risk-Beta (10, 140) to account for uncertainty around this value.

2.2.3. HERD and ANIMAL status

As [Martin et al. \(2007a\)](#) indicates, design prevalence (P^*) values define the prevalence assumption for which the sensitivity estimate is valid and are not related to any actual prevalence of disease in the population under study. Design prevalences can be set following international standards, trading partner requirements, biological plausibility or resources consideration. For diseases with a very low expected prevalence, such as the case of FMD in Australia in early stages of an outbreak, design prevalence levels are often based on what is feasible for the surveillance evaluation, instead of solely on biological grounds ([Martin et al., 2007a](#)).

The HERD STATUS node has two possible outcomes, *Infected* and *Not infected*, with the probability of a herd being infected equal to the herd-level design prevalence

Table 1

Nodes in the scenario tree representing the saleyard and abattoir surveillance components used to evaluate the sensitivity of detection of a foot-and-mouth disease outbreak at these locations in Australia.

Node	Name	Type	Sources of data	Branches
1	FARM TYPE	Category	Industry statistics (APL, 2008) Hernández-Jover et al. (2009)	Small herds Large herds
2	SWILL FEEDING	Risk category	Producers' interviews (Schembri, 2008; Schembri et al., 2010a)	Swill feeding Not swill feeding
3	HERD STATUS	Infection	Herd-level design prevalence	Infected No infected
4	ANIMAL STATUS	Infection	Unit-level design prevalence	Infected No infected
5	CLINICAL SIGNS	Detection	Kitching and Alexandersen (2002), Alexandersen et al. (2003), Orsel et al. (2009)	Displaying clinical signs Not displaying clinical signs
6	STOCKMAN DETECTION	Detection	Legislation review Stakeholders' focus group Observational studies (Schemann et al., 2010)	Detection No detection
7	INSPECTOR INSPECTION	Detection	Legislation review Stakeholders' focus group Observational studies (Schemann et al., 2010)	Inspection No inspection
8	INSPECTOR DETECTION	Detection	Legislation review Stakeholders' focus group Observational studies (Schemann et al., 2010)	Detection No detection
9	INSPECTOR PRESENCE	Detection	Legislation review Stakeholders' focus group Observational studies (Schemann et al., 2010)	Present Absent

(P_H^*). For this analysis the value of 0.01 was used as P_H^* . Like the herd infection node, the ANIMAL STATUS node had two outcomes, *Infected* and *Not infected*. The probability that the animal (i.e. pig) was infected within an infected herd was the unit-level design prevalence (P_U^*), and for this analysis 0.30 was used. To evaluate the influence of the herd and unit design prevalences on the sensitivity of detection of each component, the models were also simulated with P_H^* of 0.05 and P_U^* of 0.50 and 0.80.

2.2.4. CLINICAL SIGNS

The CLINICAL SIGNS node was included in the model as disease can only be detected in these SSCs when infected animals are displaying clinical signs, since detection at post-farm-gate venues is based on physical inspection. The CLINICAL SIGNS node had two outcomes: *Displaying clinical signs* and *Not displaying clinical signs*. The incubation period of FMD in pigs varies with the strain and the dose of the virus, the infection route, the individual susceptibility and the environment where animals are kept. For some strains, the incubation period may be as short as 24 h but more frequently clinical signs in pigs appear 2 or more days post-infection. A maximum incubation period of 11 days should be considered in pigs (Kitching and Alexandersen, 2002). Experimental studies (Alexandersen et al., 2003; Orsel et al., 2009) reported an incubation period in pigs of 1–2 days after intradermal inoculation of FMD virus. Infected pigs presented for sale or slaughter at the SSC could have been infected for at least 1 day. According to this, we assumed there would be a high probability that infected pigs would show clinical signs when presented for sale or slaughter. However, to account for the wide range of the incubation period, this probability was estimated with an @RISK function RiskPert(0.1,0.7,1.0).

2.2.5. STOCKMAN DETECTION

For this analysis, any person responsible for moving animals during unloading on arrival and transferring between pens at saleyards and abattoirs was defined as a stockman. The two possible outcomes of the STOCKMAN DETECTION node were: *Detection* and *No detection*. In all four SSCs stockmen must inspect animals to ensure animal welfare and health are not compromised, and to notify any suspicion of exotic disease to their supervisor. Therefore, stockmen could detect an infected animal showing clinical signs. This is a detection category node; both branches can lead to detection of FMD.

2.2.6. INSPECTOR INSPECTION

Inspectors at the saleyards represent health officers or stock inspectors employed by the State government. At domestic abattoirs an inspector is a quality assurance (QA) inspector employed by the abattoir, while at export abattoirs inspectors are AQIS veterinarians or authorised officers under the supervision of a veterinary officer. The outcomes of the INSPECTOR INSPECTION node were *Inspection* or *No inspection*. The probability of the inspector inspecting the animals depended on whether or not the stockman previously detected the infected animal.

If the stockman detects a clinical suspect, he/she is required to notify the inspector, who must inspect the suspect animal. Therefore, the probability that the inspector would inspect the suspect animals when a stockman had previously detected them was given a fixed value of one in all SSCs. In contrast, if the stockman had not detected a clinical suspect, inspector inspection will depend on the presence of the inspector at the SSC; thus, the probability that an inspector would inspect the animals was different among the four SSCs.

2.2.7. INSPECTOR PRESENCE

The next node in the scenario following the *No detection* branch of the STOCKMAN DETECTION node was the INSPECTOR PRESENCE node. The two outcomes for this node were *Present* or *Absent*.

2.2.8. INSPECTOR DETECTION

The possible outcomes of this node were *Detection* and *No detection*. The probability of an inspector detecting an animal infected with FMD and displaying clinical signs took a value which was conditional on stockman detection or non-detection of this animal. Thus, different probabilities for the branches of this node (*Detection* or *No detection*) were used for the different limbs of the scenario tree depending on the stockman detection. When the outcome was *Detection*, inspector notification to the relevant authorities and subsequent diagnosis of FMD was assumed and as such this node was considered a terminal node.

2.3. Parameter estimates for each surveillance system component

2.3.1. Saleyards

Approximately 5% of pigs produced in Australia are traded at livestock markets (Cutler and Holyoake, 2007). Any person in charge of animals at saleyards has responsibility for ensuring the animals are fit to sell and their welfare is not compromised, and for notifying suspicions of exotic animal diseases to the relevant authorities. However, variations in state legislation, lack of government resources for disease surveillance at saleyards, and poor training in exotic disease recognition among stockman and inspectors, have been identified. Stockman inspection at saleyards has only a 'fit for sale' purpose, and the frequency of inspectors' attendance is very low. In addition, the proportion of pigs inspected by the inspectors varies from 50 to 100%, depending on the saleyard. The ability of inspectors to detect a suspect FMD case depends on their knowledge of FMD and the purposes of their inspection. While reasons for inspection vary between states in Australia, traceability and welfare generally have priority over disease surveillance. However, knowledge of exotic diseases among government inspectors was considered to be high, as specific training is provided by the government agencies.

Branch probabilities and proportions for each of the nodes in the saleyard scenario tree are detailed in Table 2. The only difference between input values used for the domestic and export saleyards was the proportion of *Small* and *Large herds* selling animals through each of these locations.

2.3.2. Abattoirs

According to the observational studies (Schemann et al., 2010) and disease surveillance legislation review, disease surveillance activities and responsibilities differ between domestic and export abattoirs. However, all pigs destined for human consumption must be individually inspected both from the pen-side and during movement before slaughter, to assess their suitability for human consumption (CSIRO, 2007).

2.3.2.1. Domestic abattoirs. Domestic processing plants are required by the State Food Authority to have a quality assurance program that is periodically audited, for all the abattoir's processes. The QA manager or inspector, who is employed by the abattoir, is responsible for the ante-mortem inspection of pigs. We found that stockmen, who are not trained for exotic disease recognition, perform a general inspection of the animals to identify those that are not fit for slaughter. However, a high proportion of animals are unloaded at night and thus not inspected during movement. Quality assurance inspectors, although meant to perform ante-mortem inspection of all animals from the pen-side and during movement in order to identify health and welfare cases, were found to be present approximately 50% of the time, and only some animals were inspected. Inspectors are trained as meat inspectors, but not in exotic disease recognition in live animals, and thus their ability to detect a FMD case will depend on their serendipitous knowledge of FMD and their inspection priorities.

Branch probabilities and proportions used in the domestic abattoir analysis are shown in Table 3.

2.3.2.2. Export abattoirs. An AQIS inspector must conduct ante-mortem inspection of all animals (Anon., 2008). Schemann et al. (2010) found that stockmen at export abattoirs inspect all animals with the purposes of identifying welfare- and disease-compromised animals, and ensuring compliance with traceability requirements. Although stockmen at this location are not trained in exotic disease recognition, their ability to detect animals with FMD was considered higher than that of the stockmen at domestic abattoirs. The same studies reported that the AQIS inspector inspected all pigs present in the lairage from the pen-side before the start of slaughter. At the end of slaughter, the AQIS inspector also inspected animals that were previously separated as suspects by the stockman. However, pigs unloaded during the day were only inspected if the AQIS inspector was present at the lairage.

The input values used to estimate branch probabilities and proportions in the export abattoir analysis are detailed in Table 4.

2.4. Sensitivity of the surveillance system components

Sensitivity of surveillance at a single representative location for each surveillance system component for FMD was estimated for a period of two weeks, including two sale days in each saleyard and two weeks of slaughter at the abattoirs.

2.4.1. Adjusted risks

Feeding of swill was the only risk factor for herd infection considered in this model. Relative risks of *Swill feeding* and *Not swill feeding* were adjusted (Eq. (1)) to maintain their relativity while ensuring that the average risk for the population was one (Martin et al., 2007a).

$$AR_i = \frac{RR_i}{\sum_{i=1}^I RR_i \times PrP_i} \quad (1)$$

Table 2

Branch probabilities and proportions in the domestic and export saleyard scenario tree model of the likelihood of detection of a foot and mouth disease outbreak in Australia.

Node	Branch of the node	Parameter estimate ^a	Input value	Data sources
FARM TYPE	Small herds	Proportion small herds selling at saleyards	Domestic saleyard: Beta(38,4) Export saleyard: Beta(68,11)	Hernández-Jover et al. (2009)
	Large herds	Proportion large herds selling at saleyards	1-Prp of small herds	
SWILL FEEDING	Swill feeding	Proportion producers swill feeding	Small herds: Pert(0.10,0.16,0.20)	Hernández-Jover et al. (2010), Schembri (2008), Schembri et al. (2010a)
	Not swill feeding		Large herds: Pert(0.06,0.10,0.12)	
HERD STATUS	Infected	Probability herd is infected	0.01	Herd design prevalence (P_H^*)
ANIMAL STATUS	Infected	Probability pigs within an infected herd are infected	0.30	Unit design prevalence (P_U^*)
	Not infected			
CLINICAL SIGNS	Displaying clinical signs	Probability infected pig will show FMD clinical signs	Pert(0.1,0.7,1)	Kitching and Alexandersen (2002), Alexandersen et al. (2003), Orsel et al. (2009)
	Not displaying clinical signs			
STOCKMAN DETECTION	Detection	Probability stockman detects the FMD infected pigs	Pert(0.1,0.3,0.5)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	No detection			
INSPECTOR INSPECTION	Inspection	Probability inspector inspects pigs	Stockman detection limb: 1	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	No inspection		Stockman no detection limb: Pert(0.5,0.8,1)	
INSPECTOR DETECTION	Detection	Probability inspector detects the FMD infected pigs	Stockman detection limb: Pert(0.7,0.9,1)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	No detection		Stockman no detection limb: Pert(0.6,0.8,0.9)	
INSPECTOR PRESENCE	Present	Probability inspector is present at the saleyard	Pert(0.1,0.5,0.8)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	Absent			

^a FMD: foot and mouth disease.

where RR_i is the specified relative risk and AR_i the adjusted relative risk for the i th branch of the node; PrP_i is the proportion of the reference population for each branch; and I is the number of branches in the risk node.

The relative risk of swill feeding (RR_{SF}) and the proportions of small and large herds swill feeding and not swill feeding are explained in Section 2.2.2.

The adjusted risk was then used to calculate the effective probability of infection ($EPIH$) for herds *Swill feeding* and for those *Not swill feeding* (Eq. (2)).

$$EPIH_i = AR_i \times P_H^* \quad (2)$$

2.4.2. Unit sensitivity (SeU)

This is the probability that an infected pig will be detected (i.e. give a positive outcome of the SSC). It was determined from the probabilities of detection previously described (see Sections 2.2 and 2.3). For each limb of the scenario tree below the animal status node (see Fig. 1), branch probabilities were multiplied to give a limb probability, and all positive outcomes summed to obtain the SeU .

2.4.3. Herd sensitivity

In order to account for clustering of FMD in herds, surveillance sensitivity was calculated for each individual herd selling through the saleyard or the abattoir. The herd sensitivity (SeH_j) is the probability that one or more positive (detection) outcomes will be obtained when n_j pigs from the herd are processed in the surveillance system component, and the herd is infected at the unit design prevalence (Martin et al., 2007a).

In calculating SeH for each herd processed, a formula approximating the hypergeometric probability (Eq. (3)) was used for the *Small herds*, since the proportion of animals sampled from each herd was assumed higher than 10% of the herd size, thus violating the assumption of independence required when using the binomial probability formula (MacDiarmid, 1988; Martin et al., 2007a). The binomial probability formula was used to calculate SeH for *Large herds* (Eq. (4)).

$$SeH_j = 1 - \left(1 - SeU \times \frac{n_j}{N_j} \right)^{(P_U^* \times N_j)} \quad (3)$$

$$SeH_j = 1 - (1 - P_U^* \times SeU)^{n_j} \quad (4)$$

Table 3

Branch probabilities and proportions in the domestic abattoir scenario tree model of the likelihood of detection of a foot and mouth disease outbreak in pigs in Australia.

Node	Branch of the node	Parameter estimate ^a	Input value	Data sources
FARM TYPE	Small herds	Proportion small herds selling at domestic abattoirs	Beta(32,10)	Hernández-Jover et al. (2009)
	Large herds	Proportion large herds selling at domestic abattoirs	1-Prp of small herds	
SWILL FEEDING	Swill feeding	Proportion producers swill feeding	Small herds: Pert(0.10,0.16,0.20)	Hernández-Jover et al. (2010), Schembri (2008), Schembri et al. (2010a)
	Not swill feeding		Large herds: Pert(0.06,0.10,0.12)	
HERD STATUS	Infected	Probability herd is infected	0.01	Herd design prevalence (P_H^*)
	Not infected			
ANIMAL STATUS	Infected	Probability pigs within an infected herd are infected	0.30	Unit design prevalence (P_U^*)
	Not infected			
CLINICAL SIGNS	Displaying clinical signs	Probability infected pig will show FMD clinical signs	Pert(0.1,0.7,1)	Kitching and Alexandersen et al. (2002), Alexandersen et al. (2003), Orsel et al. (2009)
	Not displaying clinical signs			
STOCKMAN DETECTION	Detection	Probability stockman detects the FMD infected pigs	Pert (0.1,0.3,0.5)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	Not detection			
INSPECTOR INSPECTION	Inspection	Probability inspector inspects pigs	Stockman detection limb: 1	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	Not inspection		Stockman no detection limb: Pert(0.5,0.6,0.8)	
INSPECTOR DETECTION	Detection	Probability inspector detects the FMD infected pigs	Stockman detection limb: Pert(0.8,0.9,0.95)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	Not detection		Stockman no detection limb: Pert(0.7,0.8,0.9)	
INSPECTOR PRESENCE	Present	Probability inspector is present at the domestic abattoir	Pert(0.2,0.5,0.7)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	Absent			

^a FMD: foot and mouth disease.

where n_j is the number of pigs sold through the saleyard or abattoir for the j th herd and N_j is the total number of pigs in the j th herd.

Specific individual information about herds selling through saleyards and abattoirs, such as herd size and number of animals sold, was not available, and a data set with herd information for each SSC was simulated (see Section 2.4.4).

For a herd to be categorized as infected at least one animal had to be infected with FMD, so in Eq. (3) the minimum number of infected animals in an infected herd (otherwise $P_U^* \times N_j$) was 1.

2.4.4. Simulating the herd information

As specific information on individual herds selling through each location was not available, data sets were simulated, with numbers of records equal to the numbers of herds selling in representative locations for each of a domestic saleyard, an export saleyard, a domestic abattoir and an export abattoir, during a two-week period. Tables 5 and 6 present the @RISK functions used for the simulations and their parameters for the saleyards and the abattoirs. Information used for the herd simulation was estimated from previous research (Schembri, 2008;

Hernández-Jover et al., 2009). Estimated numbers of herds selling pigs to the domestic saleyard, export saleyard, domestic abattoir and export abattoir in a 2-week period were 20, 40, 25 and 40, respectively (Hernández-Jover et al., 2009). FARM TYPE and SWILL FEEDING were randomly allocated based on the estimated proportion of each type of farm selling through each location. The size of the *Small herds* selling through saleyards, including sows and grower pigs, was simulated based on information gathered in producer interviews (Hernández-Jover et al., 2008; Schembri, 2008). For *Large herds* selling through saleyards and all herds selling through abattoirs, size was simulated based on industry statistics (APL, 2008). The number of animals sold by each herd at all locations was simulated based on data from the saleyard and abattoir audit exercises (Section 2.1.7; Hernández-Jover et al., 2009).

2.4.5. Location sensitivity

Individual herd sensitivities (SeH_j) were used to calculate the sensitivity of the surveillance activities for a single representative location of each surveillance system component (*Location Se*). This *Location Se* is defined as the probability that one or more herds processed in each location will give a positive outcome (detection) when infection

Table 4

Branch probabilities and proportions in the export abattoir scenario tree model of the likelihood of detection of a foot and mouth disease outbreak in Australia.

Node	Branch of the node	Parameter estimate ^a	Input value	Data sources
FARM TYPE	Small herds	Proportion small herds selling at export abattoirs	Beta(65,75)	Hernández-Jover et al. (2009)
	Large herds	Proportion large herds selling at export abattoirs	1-Prp of small herds	
SWILL FEEDING	Swill feeding	Proportion producers swill feeding	Small herds: Pert(0.10,0.16,0.20)	Hernández-Jover et al. (2010), Schembri (2008), Schembri et al. (2010a)
	Not swill feeding		Large herds: Pert(0.06,0.10,0.12)	
HERD STATUS	Infected	Probability herd is infected	0.01	Herd design prevalence (P_H^*)
ANIMAL STATUS	Not infected	Probability pigs within an infected herd are infected	0.30	
CLINICAL SIGNS	Infected	Probability infected pig will show FMD clinical signs	Pert(0.1,0.7,1)	Unit design prevalence (P_U^*)
	Not infected			
STOCKMAN DETECTION	Displaying clinical signs	Probability stockman detects the FMD infected pigs	Pert(0.5,0.7,0.9)	Kitching and Alexandersen (2002), Alexandersen et al. (2003), Orsel et al. (2009)
	Not displaying clinical signs			
INSPECTOR INSPECTION	Detection	Probability inspector detects the FMD infected pigs	Pert(0.7,0.8,0.9)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	Not detection			
INSPECTOR DETECTION	Inspection	Probability inspector detects the FMD infected pigs	Pert(0.98,0.99,1)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	Not inspection			
INSPECTOR PRESENCE	Detection	Probability inspector is present at the export abattoir	Pert(0.6,0.7,0.8)	Legislation review, Stakeholders' focus groups, Schemann et al. (2010)
	Not detection			

^a FMD: foot and mouth disease.

Table 5

Description of the @RISK functions and the parameters used for the simulation of a data set of representative pig herds selling at domestic and export saleyards in Australia during a 2-week period.

Variable	@RISK function	Parameters
1. Farm type	RiskDiscrete(<i>Farm type, Frequencies</i>)	Proportions of <i>Small</i> and <i>Large</i> herds
2. Swill feeding	RiskDiscrete(<i>Swill feeding Yes/No, Frequencies</i>)	Proportions of <i>Swill feeding</i> and <i>Not swill feeding</i> herds within each FARM TYPE
3. Small herd size	RiskLognorm(<i>Number of animals, Standard Deviation</i>)	<i>Number of animals</i> : Average number of animals for small herds, 81; <i>Standard deviation</i> , 103. Truncated with a minimum (1) and maximum (1000)
3.1. Animals sold domestic saleyard	RiskLognorm(<i>Number of animals, Standard Deviation</i>)	<i>Number of animals</i> : Average number of animals for small herds, 13; <i>Standard deviation</i> , 9. Truncated with a minimum (1) and maximum (<i>Herd size-1</i>)
3.2. Animals sold export saleyard	RiskLognorm(<i>Number of animals, Standard Deviation</i>)	<i>Number of animals</i> : Average number of animals for small herds, 17; <i>Standard deviation</i> , 11. Truncated with a minimum (1) and maximum (<i>Herd size-1</i>)
4. Large herd size	RiskCumul(1000, 36287, <i>Animals by herd category, CumProp</i>)	<i>Animals by herd category</i> : Average number of animals for each herd size category (100–499 sows, $n = 2387$; 500–999 sows, $n = 6846$, and, 1000+ sows, $n = 36,287$); <i>CumProp</i> : Cumulative proportions by herd category (77.6%, 89.7% and 100%, respectively).
4.1. Animals sold domestic saleyard	RiskCumul(50, 180, <i>Bins, CumProp</i>)	<i>Bins</i> : 3 bins (70, 100, 180) according to the number of animals sold by large herds; <i>CumProp</i> : Cumulative proportion of herds in each bin (33.3%, 66.4% and 100%, respectively).
4.2. Animals sold export saleyard	RiskCumul(50, 250, <i>Bins, CumProp</i>)	<i>Bins</i> : 4 bins (60, 90, 140, 250) according to the number of animals sold by large herds; <i>CumProp</i> : Cumulative proportion of herds in each bin (20%, 60%, 90% and 100%, respectively).

Table 6

Description of the @RISK functions and the parameters used for the simulation of a data set of representative pig herds selling at domestic and export abattoirs in Australia during a 2-week period.

Variable	@RISK function	Parameters
1. Farm type	RiskDiscrete(<i>Farm type, Frequencies</i>)	Proportion farm type according to herd size
2. Swill feeding	RiskDiscrete(<i>Swill feeding Yes/No, Frequencies</i>)	Proportion <i>Swill feeding</i> and <i>Not swill feeding</i> herds within each FARM TYPE
3. Small herd size	RiskCumul(<i>10,737, Number of animals, CumProp</i>)	<i>Animals by herd category</i> : Average number of animals for each herd size category (1–49 sows, $n = 120$ and 50–99, $n = 737$); <i>CumProp</i> : Cumulative proportions by herd category (87.6% and 100%, respectively).
3.1. Animals sold domestic abattoir	RiskLognorm(<i>Number of animals, Standard Deviation</i>)	<i>Number of animals</i> : Average number of animals for small herds, 43; <i>Standard deviation</i> , 26. Truncated with a minimum (1) and maximum (<i>Herd size</i> -1).
3.2. Animals sold export abattoir	RiskLognorm(<i>Number of animals, Standard Deviation</i>)	<i>Number of animals</i> : Average number of animals for small herds, 44; <i>Standard deviation</i> , 28. Truncated with a minimum (1) and maximum (<i>Herd size</i> -1).
4. Large herd size	RiskCumul(<i>1000, 36287, Animals by herd category, CumProp</i>)	<i>Animals by herd category</i> : Average number of animals for each herd size category (100–499 sows, $n = 2387$; 500–999 sows, $n = 6846$, and, 1000+ sows, $n = 36,287$); <i>CumProp</i> : Cumulative proportions by herd category (77.6%, 89.7% and 100%, respectively).
4.1. Animals sold domestic abattoir	RiskCumul(<i>100, 700, Bins, CumProp</i>)	<i>Bins</i> : 4 bins (120, 200, 300, 700) according to the number of animals sold by large herds; <i>CumProp</i> : Cumulative proportion of herds in each bin (44.4%, 66.7%, 88.9% and 100%, respectively).
4.2. Animals sold export s abattoir	RiskLognorm(<i>Number of animals, Standard Deviation</i>)	<i>Number of animals</i> : Average number of animals for large herds, 449; <i>Standard deviation</i> , 470. Truncated with a minimum (100) and maximum (<i>Herd size</i> -1).

is present in the country at the herd and unit design prevalence (Martin et al., 2007a).

During a 2-week period, the number of herds processing pigs through a location (saleyard or abattoir) was greater than 10% of the total number of herds supplying pigs to the location. A formula approximating the hypergeometric probability was therefore used to combine the individual herd sensitivities:

$$\text{Location } Se_k = 1 - \left(1 - AveSeH_k \times \frac{h_k}{H_k}\right)^{(AveEPI_k \times H_k)} \quad (5)$$

where for the k th location, H_k is the total number of herds supplying it; and for a 2-week period $AveSeH_k$ represents the average SeH for the herds processing pigs; h_k is the number of herds processed; and $AveEPI_k$ is the average $EPIH$ for the herds processed.

The number of herds supplying pigs to each location (h_k) was estimated according to Hernández-Jover et al. (2009). The average number of herds that are regular suppliers to each location (H_k) was estimated from industry statistics and previous studies at saleyard and abattoirs (Cutler and Holyoake, 2007; Schembri, 2008; Hernández-Jover et al., 2009); large herds were more likely to have contracts with export abattoirs while small herds were more likely to process their pigs at saleyards, followed by domestic abattoirs. Estimated average numbers of herds that are regular suppliers to a domestic saleyard, export saleyard, domestic abattoir and export abattoir were 40, 70, 50 and 70, respectively.

2.4.6. Surveillance system component sensitivity

The sensitivity of each surveillance system component (*Component Se*) was calculated considering all venues in the country for each type of SSC and for the 2-week period. In Australia, there are 7 export and 19 domestic abattoirs, 3 saleyards supplying the export and domestic markets, and 10 saleyards supplying the domestic market only (Cutler and Holyoake, 2007; W. Salter, personal communication). As the *Location Se* was calculated for a representative location with typical characteristics of each type of surveillance system component, the *Component Se* for each component was obtained using the same method and equation (Eq. (5)) than those used to calculate the *Location Se*. It considered the total number of herds processed (h) through the surveillance system component in the 2-week period and the Australian pig population as the reference population (H).

2.5. Statistical analysis

Differences between *Location Se* and *Component Se* of each of the surveillance system component were statistically investigated using a non-parametric Kruskal–Wallis analysis of variance. To perform this analysis, the output values of the 3000 iterations for each *Location Se* and *Component Se*, obtained in the simulation process, were entered in Excel (PC/Windows XP, 2006) and imported into Genstat Release 9.1[®] (PC/Windows XP, 2006, VSN International Ltd., Hemel Hempstead, UK). The analysis of variance tested the null hypothesis that the

output values came from distributions with the same mean.

2.6. Sensitivity analysis

The sensitivity of the output (*Location Se*) for each surveillance component to the input variables was evaluated using @Risk's Advanced Sensitivity Analysis. The input variables evaluated were the proportion of producers *Swill feeding* for *Small* and *Large herds*, the relative risk of infection associated with *Swill feeding* (*RR.SF*), and those variables related to the probability of detection at each location (Stockman detection, Inspector inspection, Inspector presence and Inspector detection). *Location Se* was simulated using a series of fixed values for a given input variable, thus providing information on the influence of each input variable on the output of the model. Variables related to the probability of detection were allowed to vary from 0 to 1 in tenths (0.1, 0.2, 0.3 . . .); values for the proportion of producers *swill feeding* were 1%, 5%, 15% and 30%; and values for the *RR* were 2, 50, 100 and 500. Each value for each input variable was evaluated separately in a simulation of 1000 iterations, while values for all other input variables were fixed to the base value detailed in Tables 2–4.

3. Results

The median, 5 and 95 percentiles of the output distributions for *Location Se* at domestic and export saleyards and abattoirs are presented in Table 7. The *Location Se* at the export saleyard (median 0.401) was higher ($P < 0.001$) than at the domestic saleyard (median 0.190). Similarly, *Location Se* was higher ($P < 0.001$) at export abattoirs (median 0.532) than at domestic abattoirs (median 0.322). The *Component Se* for each surveillance system component in Australia is shown in Table 8. The probability that 1 or more herds will be detected in the surveillance system component during the 2-week period, given the country is infected at 1% herd design prevalence, was very high for all components. However, higher *Component Se* ($P < 0.001$) was observed at abattoirs (domestic, median 0.999; export, median 0.981) than at saleyards (domestic, median 0.847; export, median 0.710).

Table 7 also presents estimates of the *Location Se* of each component for different design prevalences (P_H^* and P_U^*). As expected the *Location Se* for all components increased considerably when increasing P_H^* , as the number of infected herds being processed at these locations would be higher. However, the effect of increasing the unit-level design prevalence on the *Location Se* was not as great as the effect of increasing herd-level design prevalence.

Results of the sensitivity analysis for each surveillance system component are shown in Fig. 2. For all surveillance system components, the input variables with most influence on the *Location Se* were the risk of FMD infection in a *Swill feeding* herd (*RR.SF*) relative to the risk in a *Not swill feeding* herd and the proportion of processed herds that were *Swill feeding*. For all surveillance system components, except the export abattoir, the proportion of *Small herds* that were *Swill feeding* had more influence on the *Location*

Se than this proportion among *Large herds*, as more *Small herds* are processed through these locations. In contrast, the proportion of *Large herds* likely to *swill feed* influenced the *Location Se* at export abattoirs to a higher degree than this proportion among *Small herds*, as more *Large herds* are processed at these locations.

For all surveillance system components, detection variables with the highest positive effect on the *Location Se* were the probabilities of stockman detection and inspector presence at the venue. At the export abattoir, variation of these probabilities had limited effect on the *Location Se*, as the ability of the stockmen to detect FMD-infected animals and the presence of the inspectors at this location were estimated higher than at the other locations.

4. Discussion

Results of this evaluation suggest that the sensitivity of detection for a single representative location for each surveillance system component during a 2-week period, is high for both saleyard and abattoir clinical surveillance. However, surveillance sensitivity at abattoirs is higher than at saleyards. The probability of FMD detection is better at abattoirs because estimates of the probabilities of the stockman and the inspector inspecting and detecting infected animals used in this evaluation were higher at these locations.

The higher probability of disease detection (*Location Se*) observed at the export saleyard compared to that at the domestic saleyard reflects the higher number of herds processing pigs at this location for the period of time evaluated, as the same input values related to the probability of detection were used for both locations. We estimated that a higher proportion of *Small herds* are *Swill feeding* than *Large herds*, so *Small herds* are higher risk herds than *Large herds*. Therefore, in those surveillance system components where more *Small herds* are processed, more infected herds will also be processed, increasing the probability of detecting disease. However, the difference in the probability of *Swill feeding* between *Small* and *Large herds* was only small. The effect of processing more small (high risk) herds is not obvious in this evaluation, since their effect on *Location Se* at locations where more of them were processed was disguised by the lower detection probabilities and lower numbers of herds processed at these locations. This can be observed when comparing the *Location Se* for the domestic and export abattoir; although a higher proportion of herds processed at the domestic abattoir are *Small herds*, the *Location Se* at this location is lower than the export abattoir *Location Se*.

When the design prevalence of FMD among herds was increased to 5%, the sensitivity of detection at saleyards and abattoirs was estimated very high. Increasing P_U^* did not have a major effect on sensitivity of detection, as detection of infection in the population mainly depends on herd status.

A national surveillance system should be able to detect a FMD outbreak as early as possible, when the prevalence among herds in the country is still low (Hadorn and Stärk, 2008). The surveillance system components evaluated in this paper form a part of the national passive

Table 7

Median (5 and 95 percentiles) of the estimated sensitivity of detection of foot and mouth disease in pigs at a representative venue of four passive surveillance system components during a 2-week period in Australia, according to the herd and unit design prevalences.^a

Surveillance Component system ^b	P_H^* 0.01 P_U^* 0.30	P_H^* 0.05 P_U^* 0.30	P_H^* 0.01 P_U^* 0.50	P_H^* 0.01 P_U^* 0.80
Domestic saleyard	0.190a (0.093:0.325)	0.651a (0.386:0.860)	0.237a (0.122:0.390)	0.274a (0.149:0.431)
Export saleyard	0.401b (0.247:0.573)	0.923b (0.760:0.986)	0.471b (0.312:0.641)	0.518b (0.356:0.682)
Domestic abattoir	0.322c (0.194:0.485)	0.857c (0.660:0.964)	0.346c (0.213:0.513)	0.356c (0.222:0.523)
Export abattoir	0.532d (0.381:0.689)	0.978d (0.909:0.997)	0.538d (0.388:0.694)	0.540d (0.389:0.696)

a,b,c,d: within a column, values without a common letter differ ($P < 0.001$).

^a P_H^* : herd design prevalence; P_U^* : unit design prevalence.

^b Domestic saleyard supplies domestic markets only; export saleyard, supplies domestic and export markets.

Table 8

Median (5 and 95 percentiles) of the estimated sensitivity of detection of foot and mouth disease in pigs at the post-farm-gate passive surveillance system components during a 2-week period in Australia, given the country is infected at 1% and 30% herd and unit design prevalences, respectively.

Surveillance system Component ^a	Number of venues	Number of herds processed	Component Se ^b
Domestic saleyard	10	200	0.847a (0.705:0.936)
Export saleyard	3	120	0.710b (0.554:0.840)
Domestic abattoir	19	475	0.999c (0.994:0.999)
Export abattoir	7	280	0.981c (0.954:0.995)

a,b,c,d: within a column, values without a common letter differ ($P < 0.001$).

^a Domestic saleyard supplies domestic markets only; export saleyard, supplies domestic and export markets.

^b Component Se: sensitivity of the surveillance system component.

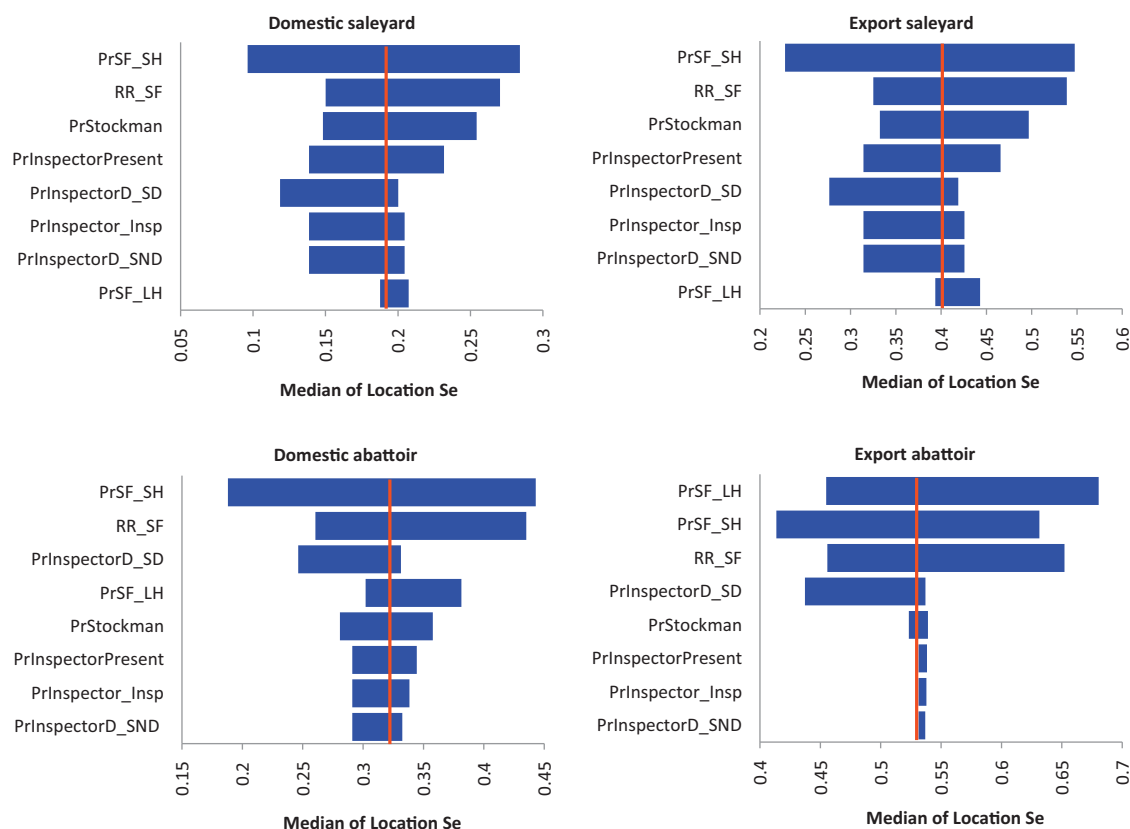


Fig. 2. Results of the sensitivity analysis representing the influence of different input variables to the median (vertical line) sensitivity of detection (*Location Se*) of foot and mouth disease at saleyard and abattoir surveillance components for pigs in Australia obtained in a simulation with 1000 iterations using @Risk's Advanced Sensitivity Analysis. (*PrSF_SH*, proportion of small herds swill feeding; *PrSF_LH*, proportion of large herds swill feeding; *PrStockman*, probability of the stockman detection; *PrInspectorPresent*, probability of inspector being present; *PrInspectorD_SD*, probability of inspector detecting following stockman detection; *PrInspector_Insp*, probability of inspector inspecting; *PrInspectorD_SND*, probability of inspector detecting following stockman non-detection; *RR_SF*, relative risk of swill feeding).

surveillance for EAD detection. In this analysis we did not intend to evaluate the whole national surveillance system for FMD detection in pigs, which would also include other passive surveillance components such as farmer or veterinarian detection and reporting, but only to evaluate the post-farm-gate components of this system. When all pig abattoirs and saleyards in Australia were considered, the probability of detecting at least one FMD-infected herd was estimated much higher than for a single location for the period of time evaluated. The greater the number of herds processed in the surveillance system component the higher the *Component Se*. The 1% herd design prevalence represents around 20 infected herds in the country, assuming there are approximately 2000 pig herds in Australia (APL, 2008). The probability of detecting at least 1 of these herds in each surveillance system component is very high (>0.700 ; Table 8). There are more abattoirs than saleyards in Australia, and more herds are processed through the abattoirs. Thus, detection at the abattoir component is higher than at saleyards and almost certain. As previously explained, 1% herd design prevalence was chosen in the model to allow for the evaluation of the surveillance system and does not represent the actual prevalence of FMD in the event of an outbreak. It is likely that if 20 pig herds were infected with FMD virus, disease would have spread to other susceptible species, in which case it could also have been detected in these species. Previous FMD outbreaks, although originating in pigs, have been detected in ruminants. Cattle are often the first species to show signs of the disease, and as such are usually called *indicators* of the disease. However, sheep and goats show only mild signs, acting as *maintenance hosts* (Centre for the Food Security and Public Health, 2007). It was not the aim of this study to evaluate either the spread of FMD from the infected piggeries or detection of disease in other susceptible species, but to identify the strengths and weaknesses of the passive post-farm-gate surveillance for pigs in Australia, which is just one part of the monitoring and surveillance system for exotic disease detection.

Passive surveillance is a cost-effective process; however, cost-intensive active surveillance could be required if the sensitivity of the surveillance system needs to be increased (Hadorn and Stärk, 2008). Although findings of this study suggest that the *Component Se* at abattoirs and saleyards is reasonably high at the specified design prevalences, some potential improvements to the system have been identified. Moreover, as detection of FMD at these locations is based on identification of clinical signs, those infected animals not showing clinical signs would not be detected. As Prattley et al. (2007) suggest, the amount of resources invested for a subgroup of the population will depend on the level of risk of introduction and the spread of a disease, and resources should be directed towards the subgroups identified as the greatest risk. One of the most likely pathways of introduction of FMD into Australia is through the illegal importation of contaminated meat which is then fed to feral or domestic pigs (Geering, 1990; Animal Health Australia, 2006). Pigs are therefore considered a greater risk for FMD introduction and spread and surveillance for this disease should focus on this species. An assessment to evaluate the magnitude of the risk of FMD

introduction and spread through small and large piggeries in Australia is currently being undertaken (Hernández-Jover et al., 2010). Saleyards and abattoirs are useful venues for conducting surveillance activities as livestock from multiple sources are assembled together. Van Metre et al. (2009) suggested that syndromic surveillance, although having a low specificity, could increase the sensitivity of detection of emerging diseases at saleyards, provided that disease syndromes are well-defined, such as a vesicular disease syndrome, which would include signs common with FMD, swine vesicular disease and vesicular stomatitis in pigs.

Data used in this evaluation to estimate the input values for the stochastic simulation model were based on different sources of information. Data were considered representative for the pig industry in the country. Information on disease surveillance activities was sourced from legislation and consultation with government and industry organizations in charge of these activities in all states in Australia and from observational studies. Uncertainty in the form of probability distributions representing the input values was included in the simulation. Output of the model is as valid as the sources of information used.

The proportion of swill feeding among small and large scale pig producers in Australia was based on feeding practices which could potentially lead to swill feeding; more detailed and precise information on swill feeding practices would have provided better estimates. As Martin et al. (2007a) indicated, the scenario tree model developed is a dynamic model which allows the ongoing evaluation of the sensitivity of the surveillance systems when new and more accurate data become available, and at the same time is a valuable tool for identifying the most efficient strategy for surveillance (Hadorn and Stärk, 2008).

According to the sensitivity analysis, the two detection probabilities with most influence on the *Location Se* are the probability of the stockman detecting disease and the probability of the inspector being present at the venue. Improving these probabilities could enhance the *Location Se* of the surveillance system component. However, at the export abattoir, as the probabilities of detection were estimated high, the *Location Se* was not greatly affected by these inputs, indicating that there is a limited potential for improving detection at this location. Schemann et al. (2010) observed that stockmen at saleyards and abattoirs inspected all pigs present at the venue and that inspector presence and inspection was less than 100%. Saleyard inspectors are required to attend at least 50% of all livestock sales in each State, including sales of other species and the method(s) of inspection for disease detection at saleyards is not specified in government legislation or state policy. The ante-mortem inspection, especially at the domestic abattoir, was inconsistent with legislative requirements, under which all pigs destined for human consumption must be individually inspected (Anon., 2008; CSIRO, 2007). Time constraints and adverse conditions at the abattoir, such as poor lighting, smell, noise and overcrowding, could potentially reduce the presence of the inspector at the lairage and the effectiveness of the inspection (Jackowiak et al., 2006; Schemann et al., 2010). Focusing on disease surveillance training of stockmen could increase the sensitivity of the

disease detection at these surveillance points. Moreover, the presence of the QA manager conducting the ante-mortem inspection of the animals should be considered crucial for enhancing the *Location Se* at domestic abattoirs. At saleyards, although an increased presence of inspectors would improve the *Location Se*, as previous studies also suggested (Schembri et al., 2006; Schemann et al., 2010), government resources available for this purpose are limited. The sensitivity analysis also indicates that the probability of the inspector detecting a FMD suspect had an influence on the *Location Se*; a decrease on this input value could decrease the *Location Se* considerably. It is important to ensure inspectors' knowledge of EAD recognition is maintained. Except at the export abattoirs, where the inspectors have a veterinary background, saleyard and domestic abattoir inspectors do not receive specific and ongoing training on EAD recognition.

The higher the *RR_SF* and the proportion of *Swill feeding* herds, the more infected herds will go through the SSC, thus, detection will be more likely to occur. However, in order to reduce the likelihood of FMD introduction into a country, the number of producers who feed swill to the pigs needs to be reduced. In Australia, this aim is achieved through different government extension campaigns to increase producer knowledge about the risks of swill feeding as well as random swill feeding inspections. Data on feeding practices of pig producers used in this study suggest that *Small herds* are more likely to swill feed than *Large herds*, potentially posing a higher risk for FMD introduction.

A risk-based surveillance approach where surveillance is focused at venues where animals pose higher risk for disease introduction could result in better allocation of disease surveillance resources (Schemann et al., 2010; Sithole et al., 2009; Stärk et al., 2006). Risk-based sampling is based on the stratification of the population into categories with different probabilities of disease introduction and spread. As Stärk et al. (2006) indicated, with reduction in available resources for surveillance, diseases and animal subpopulations that present higher risks should have higher priority for surveillance resource allocation as these targeted investments will provide higher benefit–cost ratios.

5. Conclusion

Scenario trees and stochastic modelling have been used to evaluate the sensitivity of the post-farm-gate passive surveillance system for pigs in Australia. Findings of this study suggest that sensitivity of post-farm-gate surveillance for FMD in pigs is higher at abattoirs than at saleyards. Weaknesses and potential improvements to the post-farm-gate passive surveillance system have been identified. The main recommendations from this evaluation are to improve disease awareness of saleyard and abattoir stockmen, and to increase presence of inspectors at these locations. Consideration should be given to the need for increased regulatory presence at saleyards and the potential use of a risk-based surveillance approach among small-scale pig herds.

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